

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458926.7488 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.023
Transit depth (Rp/Rs)^2	2.03 +/- 0.66 [%]
Orbital Inclination (inc)	81.56 +/- 1.72
Airmass coefficient 1 (a1)	1.099 +/- 0.012
Airmass coefficient 2 (a2)	-0.067 +/- 0.063
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	4.67
Optimal Annulus	13.3
Transit Duration (day)	0.047 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.023 vs 0.1594 (deviation 0.59σ)
Duration fit vs published	0.047 d vs 0.0512 d (deviation 0.47σ)
Mid-time O–C	-1.6 min
Depth SNR	5.1
Residual scatter	1.05 %

Light curve

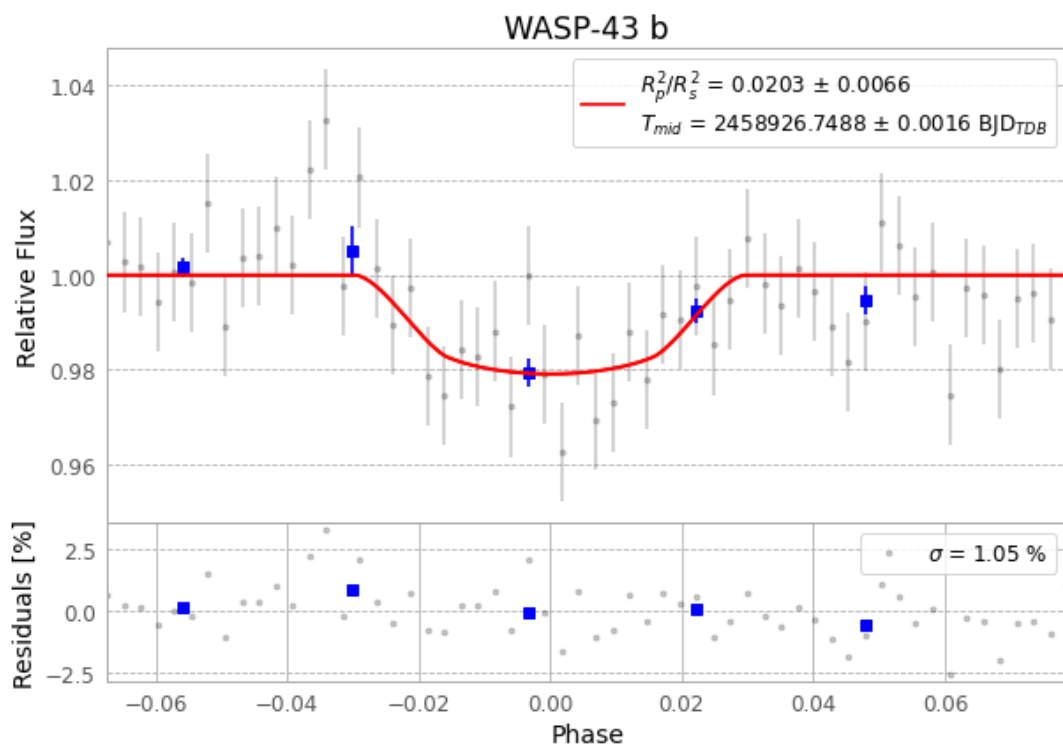


Figure 1 — FinalLightCurve_WASP-43 b_2020-03-17.png (SHA-256 d98a68d98f9e77ff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6920ad8eeca4a943...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), WASP-43200318043012.FITS → WASP-43200318072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200318.log (de0e0eeeb266...), FinalLightCurve_WASP-43 b_2020-03-17.png (d98a68d98f9e...), FinalLightCurve_WASP-43 b_2020-03-17.csv (49a272e84afe...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:43Z.